

Microduck Remaining Scenarios - End-to-End Evidence

- Microduck Remaining Scenarios: End-to-End Evidence
 - 1. Executive status
 - 2. Evidence boundaries
 - 3. Historical attempts discovered
 - 4. Repository-wide validation context
 - 5. Running: `running-v2`
 - 6. Stilt Walking: `stilts-v2`
 - 7. Swing: `swing-v1`
 - 8. Historical Running: `running-v1`
 - 9. Historical Stilt Walking: `stilts-v1`
 - 10. Stilt training failure and ELU diagnosis
 - 11. Tracked reproduction recipes
 - 12. Final conclusion

Workflow dates: September 7-8, 2026
Evidence root: `rlx/artifacts/scenarios-e2e-20260907`
Report result: **FAIL / INCOMPLETE**

1. Executive status

This report is generated from saved JSON and artifacts. It does not infer a successful skill from training completion, reward magnitude, finite output, an exported ONNX file, or a playable video. A selected scenario is complete only when its independent audit passes and its video validation passes.

Scenario	Selected attempt	Skill audit	Held-out controls	Video	E2E
Running	<code>running-v2</code>	TRAINING ACTIVE	not available	missing/fail	incomplete
Stilt Walking	<code>stilts-v2</code>	NOT RUN	not available	missing/fail	incomplete
Swing	<code>swing-v1</code>	NOT RUN	not available	missing/fail	incomplete

The aggregate result remains failed/incomplete if any selected scenario is missing or failed. This snapshot must not be described as "all verified."

2. Evidence boundaries

- Training and deterministic audit are separate. A recipe may use assisted Swing resets, but evaluation and audit start at rest.

- The effective audit environment uses the XML actuator model only. It is nominal local MuJoCo evidence, not BAM, mjlabs GPU, hardware, or sim-to-real certification.
- Stilt morphology is part of the experiment identity. The selected recipes request 2 cm morphology; reports must not substitute the API default of 10 cm.
- Fixed locomotion commands are taken from the selected recipe and reported as actual numeric values. They are not reconstructed from reward or displacement.
- The built-in recipe metrics wrapper captures the active locomotion command before `env.step()` and attaches physical metrics after the transition. This fixes the command/metric boundary when commands resample on the step boundary.
- Locomotion evaluator version 2 anchors intended heading at each twist-command change and advances it only by commanded yaw. Signed displacement therefore measures net progress along the intended command path; a closed circle cannot pass by accumulating path length while returning to its start.
- API outcome handling accepts an explicitly failed skill evaluation long enough to collect render/export evidence. A later successful render or export does not change the skill verdict.
- Outcome inference is deliberately limited: without `audit.json`, no skill verdict exists. API JSON and logs may establish TRAIN FAILED, TRAINING ACTIVE, TRAINED / UNAUDITED, or NOT RUN, but none is a pass. No pass is inferred from filenames, checkpoint existence, finite telemetry, HTTP success, screenshots, or videos.

3. Historical attempts discovered

Scenario	Attempt	Evidence status	Evidence
running	running-v1	FAIL	audit.json
running	running-v2	TRAINING LOG at 2,224,128/6,000,640	running-v2-api.log
stilts	stilts-v1	TRAIN FAILED at 1,980,416/4,001,792	stilts-v1-api.json
stilts	stilts-v2	RECIPE ONLY	stilts-v2.json
swing	swing-v1	RECIPE ONLY	swing-v1.json

Only directories containing `audit.json` carry a skill verdict.

4. Repository-wide validation context

These are saved workflow logs, not newly rerun suites. They remain relevant limitations on the evidence package:

Check	Saved result	Log
RLX native suite	297 passed, 2 skipped	rlx-native-tests.log
Broader microduck_local suite	12 failed, 384 passed, 1 skipped	microduck-contract-tests.log
Full RLX collection	collection blocked: optional mjlabs package missing	rlx-tests.log
Viewer Node tests	46 / 46 / 0	node-tests.log
Viewer build	Compiled successfully	node-build.log

Check	Saved result	Log
Viewer lint	eslint	node-lint.log

- The full RLX collection is blocked because the optional `mjlab` package is not installed in the captured environment. Native non-mjlab RLX tests passed separately.
- The broader `microduck_local` run has 12 existing failures: 11 BAM/step pre-optimization golden mismatches and one symmetry drift assertion. These failures are not hidden or reclassified by this report.

5. Running: `running-v2`

Scenario status: **TRAINING ACTIVE**

No `audit.json` exists because training is active. The retained log most recently recorded **2,224,128/6,000,640** transitions. The report remains incomplete until training, evaluation, audit, and video validation finish.

Recipe and effective audit environment

The exact requested recipe follows. These values describe training; they do not override the deterministic audit environment.

```
{
  "actionDelay": false,
  "checkpointInterval": 1000000,
  "clipCoefficient": 0.2,
  "domainRand": false,
  "entropyCoefficient": 0,
  "evalSteps": 600,
  "experimentId": "running",
  "gamma": 0.99,
  "initialStd": 0.6,
  "learningRate": 0.0003,
  "locomotionForwardCommand": 0.6,
  "maxEpisodeS": 12,
  "maxGradNorm": 0.5,
  "normalizeRewards": true,
  "numEnvs": 16,
  "numMinibatches": 4,
  "numSteps": 128,
  "obsNoise": false,
  "profile": "full",
  "randomYaw": false,
  "renderSeconds": 12,
  "resumeFromCheckpoint": false,
  "rewardWeights": {
    "air_time": 1,
    "foot_clearance": 0.5,
    "head_up": 1,

```

```
"keep_pace": 12,  
"plant_the_foot": 0.05,  
"pose": 0.1,  
"smooth_moves": 0.1,  
"stay_upright": 2  
,  
"runName": "running-e2e-20260907-v2",  
"seed": 7,  
"totalTimesteps": 6000640,  
"updateEpochs": 4  
}
```

No effective-environment object is available in the selected audit. The audit implementation fixes XML actuators, disables domain randomization, observation noise, action delay, and random yaw, and forces Swing audit starts to 0 degrees and 0 rad/s.

The requested fixed forward command is **0.6000 m/s**. If this value is *n/a*, the environment's native command sequence is evaluated rather than a fixed command.

Per-episode skill evidence

No trained-policy episode records are available.

Controls

Zero control

No records available.

Initial control

No records available.

Physics metric means

No physical metric aggregates are available.

PPO transitions and losses

No training audit summary is available.

ONNX, controls, and source identity

No ONNX or control audit is available.

API workflow

No API workflow JSON is available for this attempt.

Video and graph evidence

No `video-validation.json` is available. Images or MP4 files, if present, are linked below but are not described as browser/transport verified.

- [running-v2-api.log](#)

Source hashes

Source/artifact	Bytes	SHA-256
running.json	840	d61c43873041c403b1fb4d4bada0b40206815007660c333af4f2c9f9ddcb703d
running-v2.json	780	b20c6273b9f0f67785a3d5c9dc1f0fa685a51c541f6cb404d153714102aaf211
running-v2-api.log	5,592	7d5254cf5f51f7e6174dafff683b267da2af415ec95d7e36831bc9c63733d82a

Reproduction commands

Run from the repository root after starting the Studio server at `http://127.0.0.1:63317`:

```
node duck-viewer/scripts/rlx-dance-api-e2e.mjs --execute --base-url http://127.0.0.1:63317 --experiment
  running --recipe-json docs/remaining-scenarios-e2e/recipes/running.json --run running-reproduction-
  YYYYMMDD-HHMMSS --report rlx/artifacts/scenarios-e2e-20260907/running-reproduction-YYYYMMDD-HHMMSS-
  api.json --timeout-seconds 7200
rlx/.venv-microduck/bin/python rlx/scripts/audit_scenarios.py --recipe-json docs/remaining-scenarios-e2e/
  recipes/running.json --run rlx/runs/studio/running/running-reproduction-YYYYMMDD-HHMMSS --output
  rlx/artifacts/scenarios-e2e-20260907/running-reproduction-YYYYMMDD-HHMMSS-audit --
  seeds 101 102 103 104 105 --render
node duck-viewer/scripts/verify-rlx-video.mjs --recipe-json docs/remaining-scenarios-e2e/recipes/
  running.json --run rlx/runs/studio/running/running-reproduction-YYYYMMDD-HHMMSS --output rlx/
  artifacts/scenarios-e2e-20260907/running-reproduction-YYYYMMDD-HHMMSS-audit --base-url http://
  127.0.0.1:63317 --playwright-package "${PLAYWRIGHT_PACKAGE:-/Users/ghu/community/package.json}"
```

The audit command exits nonzero when the strict skill gate fails. That nonzero exit is expected evidence for a failed attempt, not permission to rewrite the result as successful.

6. Stilt Walking: `stilts-v2`

Scenario status: NOT RUN

No `audit.json` exists at `rlx/artifacts/scenarios-e2e-20260907/stilts-v2-audit`. The recipe may be queued. No result is inferred from a recipe file, checkpoint name, API state, or partial artifact.

Recipe and effective audit environment

The exact requested recipe follows. These values describe training; they do not override the deterministic audit environment.

```
{
  "actionDelay": false,
  "checkpointInterval": 1000000,
  "clipCoefficient": 0.2,
  "domainRand": false,
```

```

"entropyCoefficient": 0,
"evalSteps": 500,
"experimentId": "stilts",
"gamma": 0.99,
"initialStd": 0.6,
"learningRate": 0.0003,
"locomotionForwardCommand": 0.25,
"maxEpisodeS": 10,
"maxGradNorm": 0.5,
"normalizeRewards": true,
"numEnvs": 16,
"numMinibatches": 4,
"numSteps": 128,
"obsNoise": false,
"profile": "full",
"randomYaw": false,
"renderSeconds": 10,
"resumeFromCheckpoint": false,
"rewardWeights": {
  "ang_vel_xy_penalty": 0.02,
  "feet_air_time": 4,
  "head_pose": 0.1,
  "track_ang_vel": 0.5,
  "track_lin_vel": 20,
  "upright": 1
},
"runName": "stilts-e2e-20260907-v2",
"seed": 7,
"stiltBlend": 0,
"stiltHeightCm": 2,
"stiltMassKg": 0.014,
"totalTimesteps": 6000640,
"updateEpochs": 4
}

```

No effective-environment object is available in the selected audit. The audit implementation fixes XML actuators, disables domain randomization, observation noise, action delay, and random yaw, and forces Swing audit starts to 0 degrees and 0 rad/s.

The requested morphology is **2 cm** stilts, blend **0**, mass **0.0140 kg**, with a fixed forward command of **0.2500 m/s**.

Per-episode skill evidence

No trained-policy episode records are available.

Controls

Zero control

No records available.

Initial control

No records available.

Physics metric means

No physical metric aggregates are available.

PPO transitions and losses

No training audit summary is available.

ONNX, controls, and source identity

No ONNX or control audit is available.

API workflow

No API workflow JSON is available for this attempt.

Video and graph evidence

No `video-validation.json` is available. Images or MP4 files, if present, are linked below but are not described as browser/transport verified.

- No audit artifacts are present.

Source hashes

Source/artifact	Bytes	SHA-256
stilts.json	873	64bea730061e3e71fb493d4c1d78bdce17adca41929b55fa373bc56ceb68d9f4
stilts-v2.json	817	cab5e12bed6e85c866d891ec09a563f7d7cfc19a01d337a64b9354bc02a68e09

Reproduction commands

Run from the repository root after starting the Studio server at `http://127.0.0.1:63317`:

```
node duck-viewer/scripts/rlx-dance-api-e2e.mjs --execute --base-url http://127.0.0.1:63317 --experiment
  stilts --recipe-json docs/remaining-scenarios-e2e/recipes/stilts.json --run stilts-reproduction-
  YYYYMMDD-HHMMSS --report rlx/artifacts/scenarios-e2e-20260907/stilts-reproduction-YYYYMMDD-HHMMSS-
  api.json --timeout-seconds 7200
rlx/.venv-microduck/bin/python rlx/scripts/audit_scenarios.py --recipe-json docs/remaining-scenarios-e2e/
  recipes/stilts.json --run rlx/runs/studio/stilts/stilts-reproduction-YYYYMMDD-HHMMSS --output rlx/
  artifacts/scenarios-e2e-20260907/stilts-reproduction-YYYYMMDD-HHMMSS-audit --seeds 101 102 103 104
  105 --render
node duck-viewer/scripts/verify-rlx-video.mjs --recipe-json docs/remaining-scenarios-e2e/recipes/
  stilts.json --run rlx/runs/studio/stilts/stilts-reproduction-YYYYMMDD-HHMMSS --output rlx/
  artifacts/scenarios-e2e-20260907/stilts-reproduction-YYYYMMDD-HHMMSS-audit --base-url http://
  127.0.0.1:63317 --playwright-package "${PLAYWRIGHT_PACKAGE:-~/Users/ghu/community/package.json}"
```

The audit command exits nonzero when the strict skill gate fails. That nonzero exit is expected evidence for a failed attempt, not permission to rewrite the result as successful.

7. Swing: swing-v1

Scenario status: NOT RUN

No audit.json exists at rlx/artifacts/scenarios-e2e-20260907/swing-v1-audit. The recipe may be queued. No result is inferred from a recipe file, checkpoint name, API state, or partial artifact.

Recipe and effective audit environment

The exact requested recipe follows. These values describe training; they do not override the deterministic audit environment.

```
{
  "actionDelay": false,
  "checkpointInterval": 262144,
  "clipCoefficient": 0.1,
  "domainRand": false,
  "entropyCoefficient": 0.005,
  "evalSteps": 1200,
  "experimentId": "swing",
  "gamma": 0.995,
  "initialStd": 0.6,
  "learningRate": 0.0001,
  "maxEpisodes": 24,
  "maxGradNorm": 1,
  "normalizeRewards": true,
  "numEnvs": 16,
  "numMinibatches": 4,
  "numSteps": 128,
  "obsNoise": false,
  "profile": "full",
  "randomYaw": false,
  "renderSeconds": 24,
  "resumeFromCheckpoint": false,
  "runName": "swing-e2e-20260907-v1",
  "seed": 7,
  "swingInitialAngleDeg": 30,
  "swingInitialRateRadS": 1,
  "swingMinSpanDeg": 150,
  "swingPlanarActions": true,
  "totalTimesteps": 1048576,
  "updateEpochs": 3
}
```

No effective-environment object is available in the selected audit. The audit implementation fixes XML actuators, disables domain randomization, observation noise, action delay, and random yaw, and forces Swing audit starts to 0 degrees and 0 rad/s.

Training assistance is explicit in the recipe: initial angle **30 degrees**, initial rate **1 rad/s**, planar actions **yes**. The independent audit must start at rest (0 degrees, 0 rad/s); assisted training starts never count as acceptance evidence.

Per-episode skill evidence

No trained-policy episode records are available.

Controls

Zero control

No records available.

Initial control

No records available.

Physics metric means

No physical metric aggregates are available.

PPO transitions and losses

No training audit summary is available.

ONNX, controls, and source identity

No ONNX or control audit is available.

API workflow

No API workflow JSON is available for this attempt.

Video and graph evidence

No video-validation.json is available. Images or MP4 files, if present, are linked below but are not described as browser/transport verified.

- No audit artifacts are present.

Source hashes

Source/artifact	Bytes	SHA-256
swing.json	716	4b279c32981e3da6974355a50cea94a3c74a20d6306395cf66fd505c782bb125
swing-v1.json	680	1ce7e50c8a89770f89c0cb20cd5694315bf732043aac1e2ed0ddd175bd7d8f44

Reproduction commands

Run from the repository root after starting the Studio server at http://127.0.0.1:63317:

```
node duck-viewer/scripts/rlx-dance-api-e2e.mjs --execute --base-url http://127.0.0.1:63317 --experiment
swing --recipe-json docs/remaining-scenarios-e2e/recipes/swing.json --run swing-reproduction-
YYYYMMDD-HHMMSS --report rlx/artifacts/scenarios-e2e-20260907/swing-reproduction-YYYYMMDD-HHMMSS-
api.json --timeout-seconds 7200
rlx/.venv-microduck/bin/python rlx/scripts/audit_scenarios.py --recipe-json docs/remaining-scenarios-e2e/
recipes/swing.json --run rlx/runs/studio/swing/swing-reproduction-YYYYMMDD-HHMMSS --output rlx/
artifacts/scenarios-e2e-20260907/swing-reproduction-YYYYMMDD-HHMMSS-audit --seeds 101 102 103 104
105 --render
node duck-viewer/scripts/verify-rlx-video.mjs --recipe-json docs/remaining-scenarios-e2e/recipes/
swing.json --run rlx/runs/studio/swing/swing-reproduction-YYYYMMDD-HHMMSS --output rlx/artifacts/
scenarios-e2e-20260907/swing-reproduction-YYYYMMDD-HHMMSS-audit --base-url http://127.0.0.1:63317
--playwright-package "${PLAYWRIGHT_PACKAGE:-/Users/ghu/community/package.json}"
```

The audit command exits nonzero when the strict skill gate fails. That nonzero exit is expected evidence for a failed attempt, not permission to rewrite the result as successful.

8. Historical Running: running-v1

Scenario status: FAIL

Recipe and effective audit environment

The exact requested recipe follows. These values describe training; they do not override the deterministic audit environment.

```
{
  "actionDelay": false,
  "checkpointInterval": 500000,
  "clipCoefficient": 0.2,
  "domainRand": false,
  "entropyCoefficient": 0,
  "evalSteps": 600,
  "experimentId": "running",
  "gamma": 0.99,
  "initialStd": 0.35,
  "learningRate": 0.0001,
  "maxEpisodeS": 12,
  "maxGradNorm": 0.5,
  "normalizeRewards": true,
  "numEnvs": 16,
  "numMinibatches": 4,
  "numSteps": 128,
  "obsNoise": false,
  "profile": "full",
  "randomYaw": false,
  "renderSeconds": 12,
  "resumeFromCheckpoint": false,
  "runName": "running-e2e-20260907-v1",
  "seed": 7,
  "totalTimesteps": 4001792,
```

```
"updateEpochs": 4
```

```
}
```

No effective-environment object is available in the selected audit. The audit implementation fixes XML actuators, disables domain randomization, observation noise, action delay, and random yaw, and forces Swing audit starts to 0 degrees and 0 rad/s.

The requested fixed forward command is **n/a m/s**. If this value is n/a, the environment's native command sequence is evaluated rather than a fixed command.

Per-episode skill evidence

Seed	Episode	Steps	Complete	Upright	Commanded	Directed speed	Displacement	Speed ratio	Verdict
101	0	600	yes	1.0000	1.0000	0.0009	0.0119	0.0039	FAIL
102	0	600	yes	1.0000	1.0000	-0.0007	-0.0115	-0.0019	FAIL
103	0	600	yes	1.0000	1.0000	0.0012	0.0100	0.0034	FAIL
104	0	600	yes	1.0000	1.0000	-0.0013	-0.0164	-0.0040	FAIL
105	0	600	yes	1.0000	1.0000	0.0014	0.0189	0.0046	FAIL

Per-episode physics details

Seed	Episode	L/R contact	L/R air	Aerial	Support switches	L/R liftoffs	L/R touchdowns
101	0	0.9967 / 0.9967	0.0033 / 0.0033	0.0033	0	0 / 0	1 / 1
102	0	0.9967 / 0.9750	0.0033 / 0.0250	0.0033	0	0 / 13	1 / 14
103	0	0.9983 / 0.9867	0.0017 / 0.0133	0.0017	0	0 / 7	1 / 8
104	0	0.9983 / 0.9883	0.0017 / 0.0117	0.0017	0	0 / 6	1 / 7
105	0	0.9983 / 0.9867	0.0017 / 0.0133	0.0017	0	0 / 7	1 / 8

Per-episode failures

- **seed 101, episode 0:** aerial fraction below running target; both feet must lift off and touch down; command speed tracking ratio below target; command-directed displacement below target; command-directed speed below target; insufficient alternating foot support; one or both feet did not leave the ground enough
- **seed 102, episode 0:** aerial fraction below running target; both feet must lift off and touch down; command speed tracking ratio below target; command-directed displacement below target; command-directed speed below target; insufficient alternating foot support; one or both feet did not leave the ground enough

- **seed 103, episode 0:** aerial fraction below running target; both feet must lift off and touch down; command speed tracking ratio below target; command-directed displacement below target; command-directed speed below target; insufficient alternating foot support; one or both feet did not leave the ground enough
- **seed 104, episode 0:** aerial fraction below running target; both feet must lift off and touch down; command speed tracking ratio below target; command-directed displacement below target; command-directed speed below target; insufficient alternating foot support; one or both feet did not leave the ground enough
- **seed 105, episode 0:** aerial fraction below running target; both feet must lift off and touch down; command speed tracking ratio below target; command-directed displacement below target; command-directed speed below target; insufficient alternating foot support; one or both feet did not leave the ground enough

Controls

Zero control

Seed	Episode	Steps	Complete	Upright	Commanded	Directed speed	Displacement	Speed ratio	Verdict
101	0	49	no	0.7347	1.0000	-0.0875	-0.0801	-0.7074	FAIL
102	0	46	no	0.7174	1.0000	0.1081	0.0928	0.2744	FAIL
103	0	45	no	0.7111	1.0000	0.1365	0.1149	0.4550	FAIL
104	0	44	no	0.7273	1.0000	0.1296	0.1064	0.4321	FAIL
105	0	43	no	0.6977	1.0000	0.1416	0.1137	0.4721	FAIL

Physical details:

Seed	Episode	L/R contact	L/R air	Aerial	Support switches	L/R liftoffs	L/R touchdowns
101	0	0.9388 / 0.9388	0.0612 / 0.0612	0.0612	0	1 / 1	1 / 1
102	0	0.9565 / 0.9565	0.0435 / 0.0435	0.0435	0	0 / 0	1 / 1
103	0	0.9556 / 0.9778	0.0444 / 0.0222	0.0222	0	1 / 0	1 / 1
104	0	0.9773 / 0.9773	0.0227 / 0.0227	0.0227	0	0 / 0	1 / 1
105	0	0.9535 / 0.9767	0.0465 / 0.0233	0.0233	0	1 / 0	1 / 1

Failures:

- **seed 101, episode 0:** command speed tracking ratio below target; command-directed displacement below target; command-directed speed below target; episode did not complete the required 12-second horizon; episode terminated or fell; insufficient alternating foot support; one or both feet did not leave the ground enough; upright fraction below target

- **seed 102, episode 0:** both feet must lift off and touch down; command speed tracking ratio below target; command-directed displacement below target; command-directed speed below target; episode did not complete the required 12-second horizon; episode terminated or fell; insufficient alternating foot support; one or both feet did not leave the ground enough; upright fraction below target
- **seed 103, episode 0:** aerial fraction below running target; both feet must lift off and touch down; command-directed displacement below target; command-directed speed below target; episode did not complete the required 12-second horizon; episode terminated or fell; insufficient alternating foot support; one or both feet did not leave the ground enough; upright fraction below target
- **seed 104, episode 0:** aerial fraction below running target; both feet must lift off and touch down; command-directed displacement below target; command-directed speed below target; episode did not complete the required 12-second horizon; episode terminated or fell; insufficient alternating foot support; one or both feet did not leave the ground enough; upright fraction below target
- **seed 105, episode 0:** aerial fraction below running target; both feet must lift off and touch down; command-directed displacement below target; command-directed speed below target; episode did not complete the required 12-second horizon; episode terminated or fell; insufficient alternating foot support; one or both feet did not leave the ground enough; upright fraction below target

Initial control

Seed	Episode	Steps	Complete	Upright	Commanded	Directed speed	Displacement	Speed ratio	Verdict
101	0	38	no	0.7105	1.0000	-0.1217	-0.0853	-0.9839	FAIL

Physical details:

Seed	Episode	L/R contact	L/R air	Aerial	Support switches	L/R liftoffs	L/R touchdowns
101	0	0.9211 / 0.9211	0.0789 / 0.0789	0.0789	0	1 / 1	1 / 1

Failures:

- **seed 101, episode 0:** command speed tracking ratio below target; command-directed displacement below target; command-directed speed below target; episode did not complete the required 12-second horizon; episode terminated or fell; insufficient alternating foot support; one or both feet did not leave the ground enough; upright fraction below target

Physics metric means

Control	Seed	Metric	Mean
trained	101	command_forward_m_s	0.1391
trained	101	command_lateral_m_s	-0.0362
trained	101	forward_speed_m_s	0.0008

Control	Seed	Metric	Mean
trained	101	heading_forward_speed_m_s	0.0008
trained	101	heading_forward_x	0.9991
trained	101	heading_forward_y	-0.0375
trained	101	heading_lateral_speed_m_s	-0.0007
trained	101	left_foot_contact	0.9967
trained	101	right_foot_contact	0.9967
trained	101	upright	0.9994
trained	101	world_x_m	0.0036
trained	101	world_y_m	-0.0037
trained	102	command_forward_m_s	0.3497
trained	102	command_lateral_m_s	-0.0841
trained	102	forward_speed_m_s	-0.0003
trained	102	heading_forward_speed_m_s	-0.0003
trained	102	heading_forward_x	0.9741
trained	102	heading_forward_y	0.2066
trained	102	heading_lateral_speed_m_s	0.0008
trained	102	left_foot_contact	0.9967
trained	102	right_foot_contact	0.9750
trained	102	upright	0.9985
trained	102	world_x_m	-0.0019
trained	102	world_y_m	0.0117
trained	103	command_forward_m_s	0.3533
trained	103	command_lateral_m_s	0.0000
trained	103	forward_speed_m_s	0.0012
trained	103	heading_forward_speed_m_s	0.0012
trained	103	heading_forward_x	0.9886
trained	103	heading_forward_y	0.1146
trained	103	heading_lateral_speed_m_s	0.0021
trained	103	left_foot_contact	0.9983
trained	103	right_foot_contact	0.9867
trained	103	upright	0.9992
trained	103	world_x_m	0.0082
trained	103	world_y_m	0.0293

Control	Seed	Metric	Mean
trained	104	command_forward_m_s	0.2858
trained	104	command_lateral_m_s	0.0924
trained	104	forward_speed_m_s	-0.0010
trained	104	heading_forward_speed_m_s	-0.0010
trained	104	heading_forward_x	0.9872
trained	104	heading_forward_y	0.0588
trained	104	heading_lateral_speed_m_s	0.0011
trained	104	left_foot_contact	0.9983
trained	104	right_foot_contact	0.9883
trained	104	upright	0.9988
trained	104	world_x_m	0.0061
trained	104	world_y_m	0.0214
trained	105	command_forward_m_s	0.3000
trained	105	command_lateral_m_s	0.0000
trained	105	forward_speed_m_s	0.0014
trained	105	heading_forward_speed_m_s	0.0014
trained	105	heading_forward_x	0.9986
trained	105	heading_forward_y	-0.0032
trained	105	heading_lateral_speed_m_s	-0.0003
trained	105	left_foot_contact	0.9983
trained	105	right_foot_contact	0.9867
trained	105	upright	0.9993
trained	105	world_x_m	0.0019
trained	105	world_y_m	-0.0037
zero	101	command_forward_m_s	-0.0878
zero	101	command_lateral_m_s	-0.0872
zero	101	forward_speed_m_s	0.1238
zero	101	heading_forward_speed_m_s	0.1238
zero	101	heading_forward_x	0.9999
zero	101	heading_forward_y	0.0049
zero	101	heading_lateral_speed_m_s	-0.0005
zero	101	left_foot_contact	0.9388
zero	101	right_foot_contact	0.9388

Control	Seed	Metric	Mean
zero	101	upright	0.8985
zero	101	world_x_m	0.0222
zero	101	world_y_m	0.0003
zero	102	command_forward_m_s	0.3378
zero	102	command_lateral_m_s	-0.2026
zero	102	forward_speed_m_s	0.1256
zero	102	heading_forward_speed_m_s	0.1256
zero	102	heading_forward_x	0.9999
zero	102	heading_forward_y	0.0166
zero	102	heading_lateral_speed_m_s	-0.0007
zero	102	left_foot_contact	0.9565
zero	102	right_foot_contact	0.9565
zero	102	upright	0.8970
zero	102	world_x_m	0.0200
zero	102	world_y_m	-4.119e-05
zero	103	command_forward_m_s	0.3000
zero	103	command_lateral_m_s	0.0000
zero	103	forward_speed_m_s	0.1365
zero	103	heading_forward_speed_m_s	0.1365
zero	103	heading_forward_x	1.0000
zero	103	heading_forward_y	0.0004
zero	103	heading_lateral_speed_m_s	-0.0011
zero	103	left_foot_contact	0.9556
zero	103	right_foot_contact	0.9778
zero	103	upright	0.8895
zero	103	world_x_m	0.0259
zero	103	world_y_m	-0.0008
zero	104	command_forward_m_s	0.3000
zero	104	command_lateral_m_s	0.0000
zero	104	forward_speed_m_s	0.1296
zero	104	heading_forward_speed_m_s	0.1296
zero	104	heading_forward_x	0.9999
zero	104	heading_forward_y	-0.0123

Control	Seed	Metric	Mean
zero	104	heading_lateral_speed_m_s	-0.0017
zero	104	left_foot_contact	0.9773
zero	104	right_foot_contact	0.9773
zero	104	upright	0.8980
zero	104	world_x_m	0.0227
zero	104	world_y_m	-0.0010
zero	105	command_forward_m_s	0.3000
zero	105	command_lateral_m_s	0.0000
zero	105	forward_speed_m_s	0.1416
zero	105	heading_forward_speed_m_s	0.1416
zero	105	heading_forward_x	1.0000
zero	105	heading_forward_y	-0.0032
zero	105	heading_lateral_speed_m_s	0.0024
zero	105	left_foot_contact	0.9535
zero	105	right_foot_contact	0.9767
zero	105	upright	0.8880
zero	105	world_x_m	0.0262
zero	105	world_y_m	0.0016
initial	101	command_forward_m_s	-0.0878
initial	101	command_lateral_m_s	-0.0872
initial	101	forward_speed_m_s	0.1584
initial	101	heading_forward_speed_m_s	0.1584
initial	101	heading_forward_x	0.9989
initial	101	heading_forward_y	-0.0273
initial	101	heading_lateral_speed_m_s	0.0133
initial	101	left_foot_contact	0.9211
initial	101	right_foot_contact	0.9211
initial	101	upright	0.8888
initial	101	world_x_m	0.0250
initial	101	world_y_m	0.0015

PPO transitions and losses

Training evidence	Value
Finite optimizer metrics	yes
PPO rollout/update cycles	1,954
Optimizer minibatches	31,264
Observed transitions	4,001,792
Requested transitions	4,001,792
Monotonic update steps	yes
Training evidence gate	PASS
Maximum approximate KL	0.3795
Last-100 mean_loss	-0.0208
Last-100 policy_loss	-0.0236
Last-100 value_loss	0.0028
Last-100 entropy	-2.9254
Last-100 approximate_kl	0.0217
Last-100 clip_fraction	0.2639
Last-100 explained_variance	0.9255

Transitions	Steps	Raw return	Skill	Failures
0	38	215.07 71	FAIL	command speed tracking ratio below target; command-directed displacement below target; command-directed speed below target; episode did not complete the required 12-second horizon; episode terminated or fell; insufficient alternating foot support; one or both feet did not leave the ground enough; upright fraction below target
1,001,472	600	5,427.5 8	FAIL	aerial fraction below running target; both feet must lift off and touch down; command speed tracking ratio below target; command-directed displacement below target; command-directed speed below target; insufficient alternating foot support; one or both feet did not leave the ground enough
2,000,896	600	5,259.1 0	FAIL	aerial fraction below running target; both feet must lift off and touch down; command speed tracking ratio below target; command-directed displacement below target; command-directed speed below target; insufficient alternating foot support; one or both feet did not leave the ground enough
3,000,320	600	5,506.9 2	FAIL	aerial fraction below running target; both feet must lift off and touch down; command speed tracking ratio below target; command-directed displacement below target; command-directed speed below target; insufficient alternating foot support; one or both feet did not leave the ground enough

Transitions	Steps	Raw return	Skill	Failures
4,001,792	600	5,520.36	FAIL	aerial fraction below running target; both feet must lift off and touch down; command speed tracking ratio below target; command-directed displacement below target; command-directed speed below target; insufficient alternating foot support; one or both feet did not leave the ground enough

ONNX, controls, and source identity

Check	Value
Policy SHA-256	35897dbac13f06163da7add449325c6f5859023b0882bebbb6940bfb122b56b3
ONNX input	batch x 61
ONNX output	batch x 14
ONNX contract	PASS
Native/ONNX max absolute error	3.576e-07
Parameter L2 change	14.1883
Negative controls failed as required	yes
Audit scope	Nominal local simulation only; no hardware certification. No assisted Swing starts.

API workflow

Action	Accepted	Phase	Exit code	Finished
eval	yes	failed	2	2026-09-08T00:12:59.817Z
render	yes	succeeded	0	2026-09-08T00:13:10.249Z
export	yes	succeeded	0	2026-09-08T00:13:15.470Z

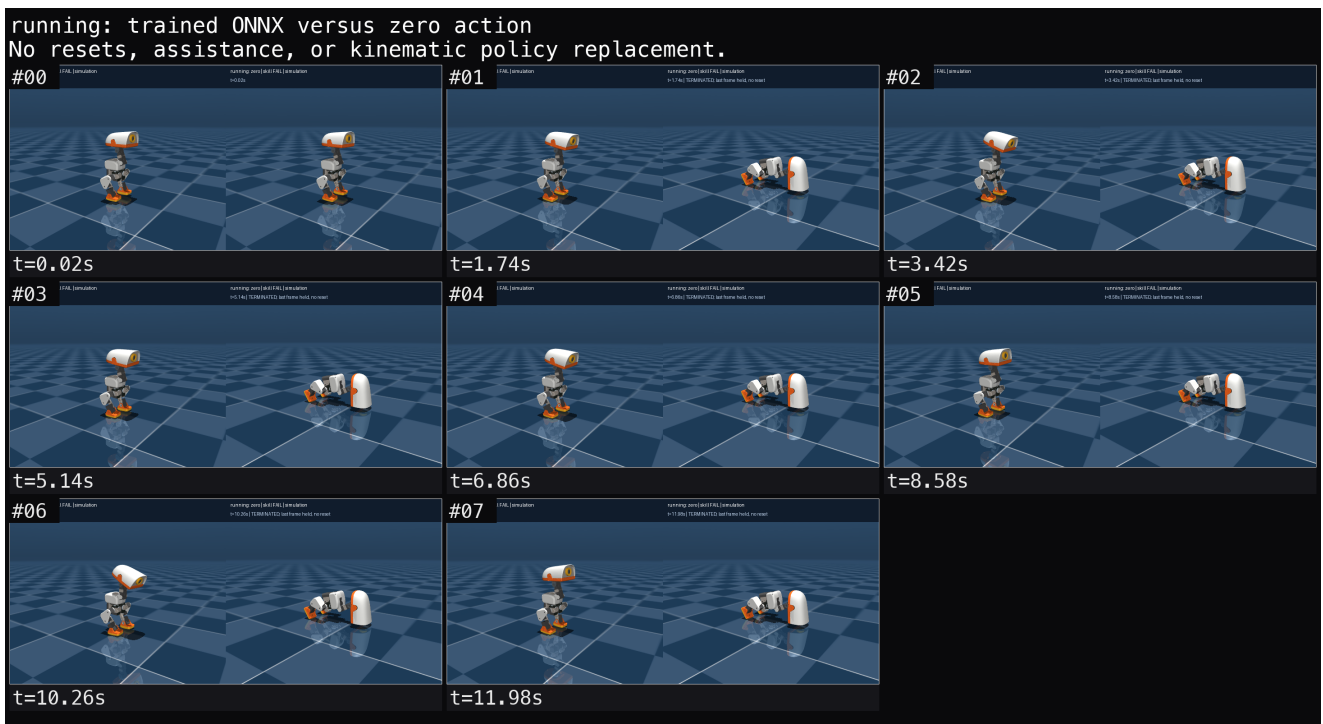
Recorded API failure: **Full Running skill evaluation failed; render and export evidence were collected.**

Video and graph evidence

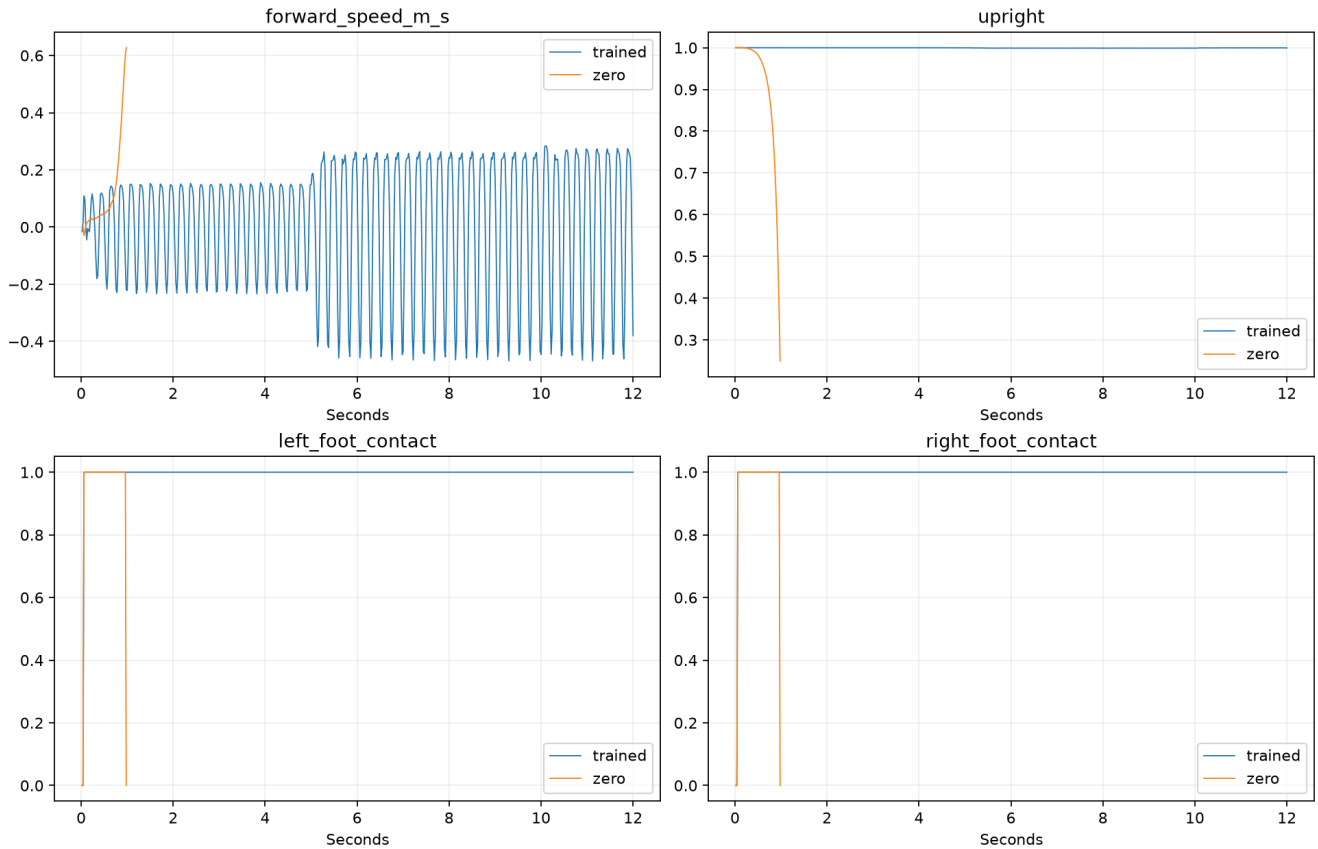
Video validation	Value
Validation status	PASS
API bytes equal local render	yes
HTTP byte range	yes
Browser playback	{"currentTime": 6, "duration": 12, "error": null, "height": 360, "width": 640}

File	Codec	Dimensions	FPS	Frames	Seconds	SHA-256
api-video.mp4	h264	640 x 360	25/ 1	300	12.00000 0	6b334ebb05e286a07566846a6485f7bfbdd6489d3db42d75d2adffad3dd156da
comparison.mp4	h264	960 x 412	25/ 1	300	12.00000 0	3c1a807fe6e7d319cb96ef5cc0cef61419f818a1aa7de655b2d69992e4586e3b

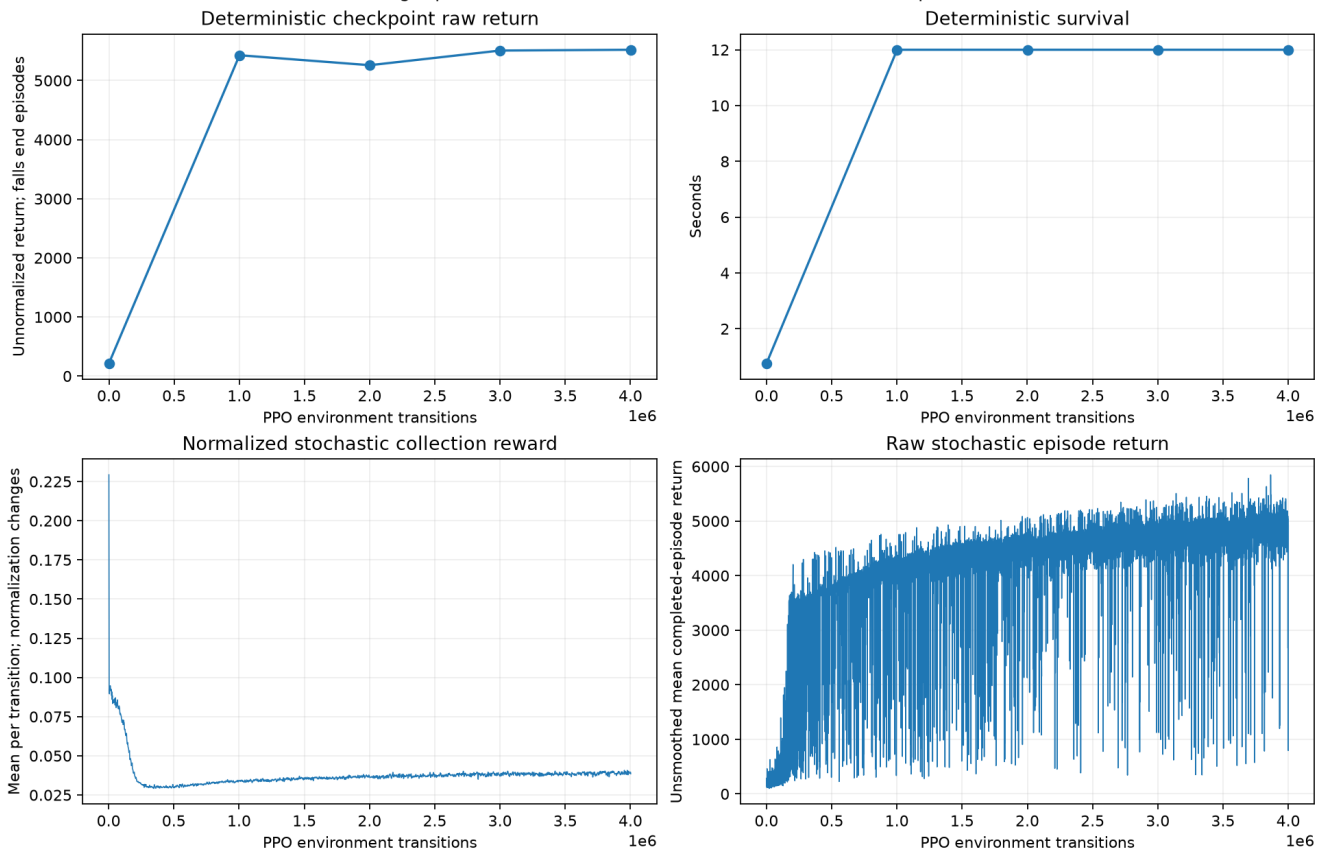
- comparison_sheet.png
- physical-tracking.png
- reward-learning.png
- ppo-losses.png
- browser-video.png
- comparison.mp4
- api-video.mp4
- audit.json
- video-validation.json
- running-v1-api.json



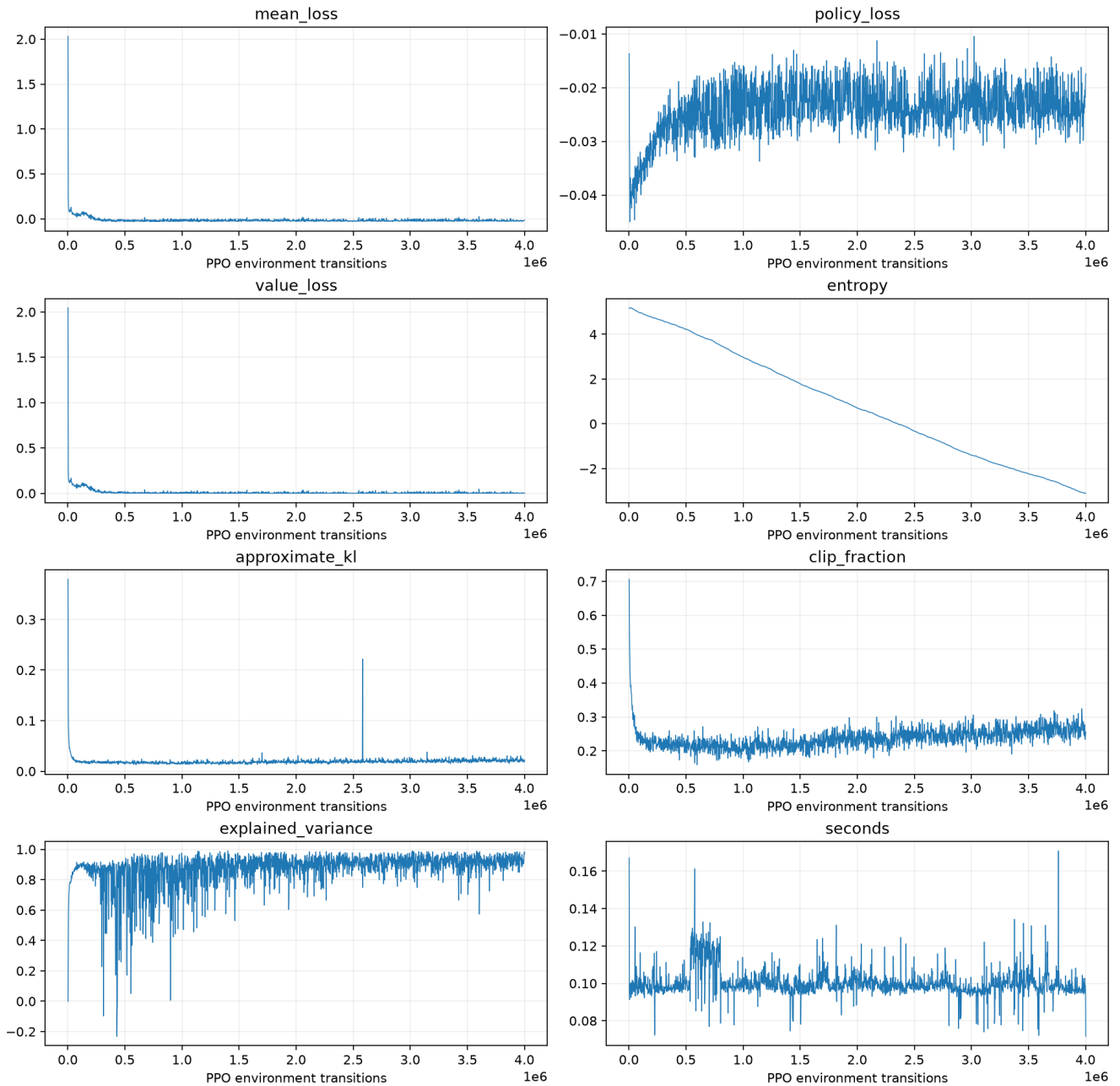
running: held-out seed physical measurements, PPO versus zero action



running: optimization evidence, not a substitute for skill acceptance



Actual PPO minibatch aggregates; full history including startup spikes



Source hashes

Source/artifact	Bytes	SHA-256
running-v1.json	570	ad3f470d5492a874c3acb652ad38ac71196ea02c20cd92e2b124290aba87567b
audit.json	72,213	481e327a90ee9794e0800b025f25d796aed698ffac6cd7931b1372f7a84b5c09
video-validation.json	6,215	7077ca733ad7c6eea6e3f195ced51f279d1487192a90838c9a8d4bd59dde86d
running-v1-api.json	307,396	a415b4196365b10d560799f26f8165f5769c4e72ae110ebbc30bbd707c1ad544
comparison_sheet.png	908,199	41f380f327930207ea70eda75ec581f904d4f0c4a6d7aeaa04cae83a9d223c15
physical-tracking.png	199,917	2b13405b65fa5066632329ad36086e5296c20f1083e5dff0358d918b949e256b
reward-learning.png	217,715	5fc2c4835af8a53380e1a788ad9844735450c82005af0cce1f8bcc5d7263a8c

Source/artifact	Bytes	SHA-256
ppo-losses.png	370,488	3098b8569940697204be6330685ae42310c559e9a2a5967d91424ba5fa4d9d35
browser-video.png	124,963	d14cbd589ddf15608f2c508651d1d3f4c39168613c0c4d10b4993a024375e131
comparison.mp4	425,184	3c1a807fe6e7d319cb96ef5cc0cef61419f818a1aa7de655b2d69992e4586e3b
api-video.mp4	386,402	6b334ebb05e286a07566846a6485f7bfbdd6489d3db42d75d2adffad3dd156da

Reproduction commands

Run from the repository root after starting the Studio server at <http://127.0.0.1:63317>:

```
node duck-viewer/scripts/rlx-dance-api-e2e.mjs --execute --base-url http://127.0.0.1:63317 --experiment
  running --recipe-json rlx/artifacts/scenarios-e2e-20260907/running-v1.json --run running-
  reproduction-YYYYMMDD-HHMMSS --report rlx/artifacts/scenarios-e2e-20260907/running-reproduction-
  YYYYYMMDD-HHMMSS-api.json --timeout-seconds 7200
rlx/.venv-microduck/bin/python rlx/scripts/audit_scenarios.py --recipe-json rlx/artifacts/scenarios-
  e2e-20260907/running-v1.json --run rlx/runs/studio/running/running-reproduction-YYYYMMDD-HHMMSS --
  output rlx/artifacts/scenarios-e2e-20260907/running-reproduction-YYYYMMDD-HHMMSS-audit --seeds 101
  102 103 104 105 --render
node duck-viewer/scripts/verify-rlx-video.mjs --recipe-json rlx/artifacts/scenarios-e2e-20260907/running-
  v1.json --run rlx/runs/studio/running/running-reproduction-YYYYMMDD-HHMMSS --output rlx/artifacts/
  scenarios-e2e-20260907/running-reproduction-YYYYMMDD-HHMMSS-audit --base-url http://
  127.0.0.1:63317 --playwright-package "${PLAYWRIGHT_PACKAGE:-/Users/ghu/community/package.json}"
```

The audit command exits nonzero when the strict skill gate fails. That nonzero exit is expected evidence for a failed attempt, not permission to rewrite the result as successful.

This historical command uses the retained artifact recipe `rlx/artifacts/scenarios-e2e-20260907/running-v1.json` because it does not match the tracked selected `recipes/running.json`. It is not a fresh-checkout reproduction unless that historical artifact recipe is separately retained.

9. Historical Stilt Walking: stilts-v1

Scenario status: TRAIN FAILED

No `audit.json` exists because training failed before a final checkpoint and deterministic audit were produced.

Training failure evidence	Value
Transitions reached	1,980,416
Requested transitions	4,001,792
Exit code	1
Started	2026-09-08T00:13:16.008Z
Finished	2026-09-08T00:18:21.415Z
Final checkpoint present	no
Final ONNX present	no

Recorded failure: RuntimeError: PPO gradients contains non-finite values: gradient tensors=critic.layers.0.weight, critic.layers.0.bias, critic.layers.2.weight, critic.layers.2.bias; observations[-7.21088..10; nonfinite=0], actions[-1.39449..3.84149; nonfinite=0], old_log_probabilities[-11.5659..8.13558; nonfinite=0], old_values[-1.64562..3.35344; nonfinite=0], advantages[-1.87452..1.62534; nonfinite=0], returns[-0.430656..3.33284; nonfinite=0]

Recipe and effective audit environment

The exact requested recipe follows. These values describe training; they do not override the deterministic audit environment.

```
{
  "actionDelay": false,
  "checkpointInterval": 500000,
  "clipCoefficient": 0.2,
  "domainRand": false,
  "entropyCoefficient": 0,
  "evalSteps": 500,
  "experimentId": "stilts",
  "gamma": 0.99,
  "initialStd": 0.25,
  "learningRate": 0.0001,
  "locomotionForwardCommand": 0.15,
  "maxEpisodeS": 10,
  "maxGradNorm": 0.5,
  "normalizeRewards": true,
  "numEnvs": 16,
  "numMinibatches": 4,
  "numSteps": 128,
  "obsNoise": false,
  "profile": "full",
  "randomYaw": false,
  "renderSeconds": 10,
  "resumeFromCheckpoint": false,
  "rewardWeights": {
    "feet_air_time": 3,
    "track_lin_vel": 10,
    "upright": 2
  },
  "runName": "stilts-e2e-20260907-v1",
  "seed": 7,
  "stiltBlend": 0,
  "stiltHeightCm": 2,
  "stiltMassKg": 0.014,
  "totalTimesteps": 4001792,
  "updateEpochs": 4
}
```

No effective-environment object is available in the selected audit. The audit implementation fixes XML actuators, disables domain randomization, observation noise, action delay, and random yaw, and forces Swing audit starts to 0 degrees and 0 rad/s.

The requested morphology is **2 cm** stilts, blend **0**, mass **0.0140 kg**, with a fixed forward command of **0.1500 m/s**.

Per-episode skill evidence

No trained-policy episode records are available.

Controls

Zero control

No records available.

Initial control

No records available.

Physics metric means

No physical metric aggregates are available.

PPO transitions and losses

No training audit summary is available.

ONNX, controls, and source identity

No ONNX or control audit is available.

API workflow

Action	Accepted	Phase	Exit code	Finished
train	yes	failed	1	2026-09-08T00:18:21.415Z

Recorded API failure: **train ended as phase=failed operation=train; last log=RuntimeError: PPO gradients contains non-finite values: gradient tensors=critic.layers.0.weight, critic.layers.0.bias, critic.layers.2.weight, critic.layers.2.bias; observations[-7.21088..10; nonfinite=0], actions[-1.39449..3.84149; nonfinite=0], old_log_probabilities[-11.5659..8.13558; nonfinite=0], old_values[-1.64562..3.35344; nonfinite=0], advantages[-1.87452..1.62534; nonfinite=0], returns[-0.430656..3.33284; nonfinite=0]**

Video and graph evidence

No `video-validation.json` is available. Images or MP4 files, if present, are linked below but are not described as browser/transport verified.

- [stilts-v1-api.json](#)

Source hashes

Source/artifact	Bytes	SHA-256
stilts-v1.json	741	25507ddb7074919a4143f3c625f1178caeed92f19ec01fa8d63279c549d3a749
stilts-v1-api.json	135,906	d01e4a21f56b0a499365a6bdad0fbc58bf5bfcc7ab1697a40002e2c995c2129e

Reproduction commands

Run from the repository root after starting the Studio server at `http://127.0.0.1:63317`:

```
node duck-viewer/scripts/rlx-dance-api-e2e.mjs --execute --base-url http://127.0.0.1:63317 --experiment
  stilts --recipe-json rlx/artifacts/scenarios-e2e-20260907/stilts-v1.json --run stilts-reproduction-
  YYYYMMDD-HHMMSS --report rlx/artifacts/scenarios-e2e-20260907/stilts-reproduction-YYYYMMDD-HHMMSS-
  api.json --timeout-seconds 7200
rlx/.venv-microduck/bin/python rlx/scripts/audit_scenarios.py --recipe-json rlx/artifacts/scenarios-
e2e-20260907/stilts-v1.json --run rlx/runs/studio/stilts/stilts-reproduction-YYYYMMDD-HHMMSS --
output rlx/artifacts/scenarios-e2e-20260907/stilts-reproduction-YYYYMMDD-HHMMSS-audit --seeds 101
102 103 104 105 --render
node duck-viewer/scripts/verify-rlx-video.mjs --recipe-json rlx/artifacts/scenarios-e2e-20260907/stilts-
v1.json --run rlx/runs/studio/stilts/stilts-reproduction-YYYYMMDD-HHMMSS --output rlx/artifacts/
scenarios-e2e-20260907/stilts-reproduction-YYYYMMDD-HHMMSS-audit --base-url http://127.0.0.1:63317
--playwright-package "${PLAYWRIGHT_PACKAGE:-/Users/ghu/community/package.json}"
```

The audit command exits nonzero when the strict skill gate fails. That nonzero exit is expected evidence for a failed attempt, not permission to rewrite the result as successful.

This historical command uses the retained artifact recipe `rlx/artifacts/scenarios-e2e-20260907/stilts-v1.json` because it does not match the tracked selected `recipes/stilts.json`. It is not a fresh-checkout reproduction unless that historical artifact recipe is separately retained.

10. Stilt training failure and ELU diagnosis

The retained `stilts-v1` API record establishes the observed failure: PPO aborted at 1,980,416 of 4,001,792 transitions because critic-layer gradients became non-finite. The logged observations, actions, old log probabilities, old values, advantages, and returns were finite, and no final checkpoint or ONNX artifact was produced.

A separate installed-MLX activation probe supplies a plausible mechanism. The stock `mlx.nn.elu` produced finite forward outputs for large positive inputs 90 and 1000 but non-finite gradients. This is consistent with the critic-only gradient failure because actor and critic use the same hidden activation structure while receiving different objectives and gradient paths. It is **not proof of the exact failing minibatch**, which was not saved.

The local model now uses a stable ELU expression:

```
where(x >= 0, x, exp(min(x, 0)) - 1)
```

This keeps the same 512-256-128 hidden-layer topology, checkpoint tensor layout, and ELU forward behavior used by exported ONNX graphs. The PPO finite-value guards remain fail-closed: a non-finite loss, gradient, or model parameter still raises; bad updates are not silently skipped.

Probe field	Captured value
Inputs	[-1.0, 0.0, 1.0, 90.0, 1000.0]
Outputs	[-0.6321205496788025, 0.0, 1.0, 90.0, 1000.0]
Gradients	[0.3678794503211975, 1.0, 1.0, null, null]
Finite outputs	yes
Finite gradients	no
Implementation	installed mlx.nn.elu: where(x > 0, x, alpha * (exp(x) - 1))

Validation artifact	Result	Link
Activation test before fix	3 failed because <code>stable_elu</code> did not yet exist	activation-before.log
Activation test after fix	3 passed: eager/compiled large-input gradient boundary and forward parity	activation-after.log
Installed MLX probe	finite forward, non-finite gradients at large positive inputs	activation-probe.json

Source hashes:

Artifact	Bytes	SHA-256
activation-probe.json	375	0074fd2183fd765c729d5746162adc302e62bd84dd0755149d0dd682830fbbe5
activation-before.log	2,068	0dab9d24d3d427ed44db37ea020c3680b9cac27c1c77ed79883208389d0b04f2
activation-after.log	98	fc99c56adf2cb5702a4564ceead9e104cf6db01c3cd1a15756b5c9630e644ce6
activation_probe.py	769	c1a52763100be1cdf44f16da9def46a54964fc10c1445999d78d38066c9bd03c

11. Tracked reproduction recipes

These copies are tracked with the report so a fresh checkout does not depend on the gitignored evidence directory:

- [Running](#)
- [Stilt Walking](#)
- [Swing](#)

12. Final conclusion

Report result: FAIL / INCOMPLETE.

The evidence package is not fully verified. Outstanding selected scenarios: Running (training active), Stilt Walking (not run), Swing (not run). Regenerate this report after their audit and video-validation artifacts exist; the generator will update the verdict from the files rather than from expectations.